Compact positive-definite spaces can have infinite magnitude

Literature-status note for `fa_banach_001`

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Status. `literature_already_answered (negative answer)`. No new mathematical result is claimed.

Original question

For a compact positive-definite metric space X, Meckes defines magnitude by the supremum of the finite quadratic ratios in display (1) of [1]. Immediately after that display, on arXiv PDF page 2, he asks whether this supremum is finite for every compact positive-definite metric space.

Explicit later answer

Leinster and Meckes explicitly identify this question on page 2 of [2]. They state that no compact positive-definite space of infinite magnitude had previously been known, cite [1] among the papers in which the question was raised, and announce examples with infinite magnitude at every positive scale.

Their Theorem 2.1, on arXiv PDF page 3, gives the answer. If (a_i) is a positive null sequence with $\sum_i a_i = \infty$, and X is the closed convex hull of $\{a_i e_i : i \geq 1\}$ in ℓ_1, then X is compact and of negative type, and

$$\operatorname{Mag}(tX) = \infty \qquad (t > 0).$$

Negative type means precisely that every positive rescaling is positive definite. In particular, X itself is compact and positive definite, so the theorem is an exact negative answer to the source question, with the stronger all-scales conclusion.

A short star-subspace verification

There is also a direct nonconvex witness inside the same ambient space. Put

$$S = \{0\} \cup \{x_n : n \geq 1\} \subset \ell_1, \qquad x_n = \frac{1}{n} e_n.$$

Then $x_n \to 0$, so S is compact. Since ℓ_1 is of negative type, S is positive definite. For $S_N = \{0, x_1, \ldots, x_N\}$, write $a_i = 1/i$ and $q_i = e^{-a_i}$. The similarity matrix $Z = (e^{-d(x,y)})_{x,y \in S_N}$ satisfies

$$Z_{0i} = q_i, \qquad Z_{ij} = q_i q_j \quad (i \neq j), \qquad Z_{ii} = 1.$$

Solving $Zw = \mathbf{1}$, set $T = w_0 + \sum_{j=1}^{N} q_j w_j$. The zeroth row gives $T = 1$, while row i gives

$$q_i T + (1 - q_i^2) w_i = 1, \qquad \text{hence} \qquad w_i = \frac{1}{1 + q_i}.$$

Consequently,

$$\mathrm{Mag}(S_N) = \sum_{i=0}^{N} w_i = 1 + \sum_{i=1}^{N} \frac{1-q_i}{1+q_i} = 1 + \sum_{i=1}^{N} \tanh\left(\frac{1}{2i}\right).$$

The last sum diverges with N. Since compact positive-definite magnitude is the supremum of the magnitudes of finite subspaces, $\mathrm{Mag}(S) = \infty$. Replacing $1/i$ by t/i proves $\mathrm{Mag}(tS) = \infty$ for every $t > 0$.

This elementary calculation independently verifies the negative answer, but it does not change the provenance classification: the exact question was already answered explicitly by [2].

Scope and search status

The identification was checked in the cached arXiv sources and PDFs. The source question is on page 2 of arXiv:1904.08923. The answering paper names the prior question on page 2 and proves its stronger Theorem 2.1 on page 3. The run's cheap indexes also contain a prior triage record for arXiv:2112.12889; that record correctly treats the later paper's own introductory signal as a same-paper ask-and-answer extraction. It does not negate the two-paper literature relation recorded here.

Other conjectures in arXiv:1904.08923, including the small-scale derivative formula for convex bodies and the Lipschitz-distance continuity conjecture, are outside this packet and are not claimed resolved.

References

[1] M. W. Meckes, *On the magnitude and intrinsic volumes of a convex body in Euclidean space*, Mathematika **66** (2020), 343–355; arXiv:1904.08923, https://arxiv.org/abs/1904.08923.

[2] T. Leinster and M. W. Meckes, *Spaces of extremal magnitude*, arXiv:2112.12889, https://arxiv.org/abs/2112.12889.